1) a)
$\mathbf{BA}$

1) b)
$\mathbf{AC}$

1) c)
$\mathbf{CB}$

1) d)
$\mathbf{BCA}$

1) e)
$\mathbf{CAB}$

2) a)
$\mathbf{BA} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix} = \begin{pmatrix} 8 & 6 \\ 2 & -3 \end{pmatrix}$

2) b)
$\mathbf{AB} = \begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 6 & 8 \end{pmatrix}$

3)
Reflection in the x-axis, $\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in the y-axis, $\mathbf{N} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
$\mathbf{NM} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
This is a rotation about the origin by $180°$

4) a)
Reflection in the line $y = x$

4) b)
$\mathbf{A}^2 = \mathbf{I}$, $\mathbf{A}^4 = \mathbf{I}$, $\mathbf{A}^6 = \mathbf{I}$, …
So $\mathbf{A}^{2024} = \mathbf{I}$, so every point maps to itself.

5) a)
Rotation, anticlockwise about the origin, by $90°$

5) b)
$\mathbf{A}^2$ is a rotation about the origin by $180°$
$\mathbf{A}^3$ is a rotation, anticlockwise about the origin, by $270°$
$\mathbf{A}^4 = \mathbf{I}$
$\mathbf{A}^5$ is a rotation, anticlockwise about the origin, by $90°$
$\mathbf{A}^6$ is a rotation about the origin by $180°$
$\mathbf{A}^7$ is a rotation, anticlockwise about the origin, by $270°$
$\mathbf{A}^8 = \mathbf{I}$
…
$\frac{335}{4} = 83$ r 3
So $\mathbf{A}^{335}$ is a rotation, anticlockwise about the origin, by $270°$

6) a)
A stretch, parallel to the x-axis, scale factor 2
Followed by
A stretch, parallel to the y-axis, scale factor 5
(either order)
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$

6) b)
An enlargement, centred at the origin, scale factor 3
Followed by
A reflection in the line $y = x$
(either order)
$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$

6) c)
An enlargement, centred at the origin, scale factor 2
Followed by
A rotation, anticlockwise about the origin, by $90°$,
(either order)
$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$

6) d)

An enlargement, centred at the origin, scale factor 5
Followed by
A reflection in the x-axis
(either order)

$$\begin{pmatrix}5 & 0\\ 0 & 5\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}=\begin{pmatrix}5 & 0\\ 0 & -5\end{pmatrix}$$

6) e)

An enlargement, centred at the origin, scale factor 4
Followed by
A rotation, about the origin, by $180°$,
(either order)

$$\begin{pmatrix}4 & 0\\ 0 & 4\end{pmatrix}\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}=\begin{pmatrix}-4 & 0\\ 0 & -4\end{pmatrix}$$

6) f)

An enlargement, centred at the origin, scale factor 8
Followed by
A reflection in the y-axis
(either order)

$$\begin{pmatrix}8 & 0\\ 0 & 8\end{pmatrix}\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}-8 & 0\\ 0 & 8\end{pmatrix}$$

6) g)

A stretch, parallel to the y-axis, scale factor 9
Followed by
A rotation, anticlockwise about the origin, by $270°$ (or a rotation, clockwise about the origin, by $90°$),
(either order)

$$\begin{pmatrix}1 & 0\\ 0 & 9\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}=\begin{pmatrix}0 & 1\\ -9 & 0\end{pmatrix}$$

7)

$\det\mathbf{A}=1\times 2-k\times 3=2-3k$
$\det\mathbf{B}=5$
If both area and orientation are preserved, then $\det\mathbf{AB}=1$
$\det\mathbf{AB}=\det\mathbf{A}\times\det\mathbf{B}\Rightarrow 1=(2-3k)\times 5\Rightarrow k=\frac{3}{5}$

8)

$\det \mathbf{A} = (2k+1) \times (-2) - 4 \times 3k = -4k - 2 - 12k = -16k - 2$

$\det \mathbf{B} = -3$

If area is preserved but not orientation, then $\det \mathbf{AB} = -1$

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} \Rightarrow -1 = (-16k - 2) \times (-3) \Rightarrow k = -\frac{7}{48}$

9)

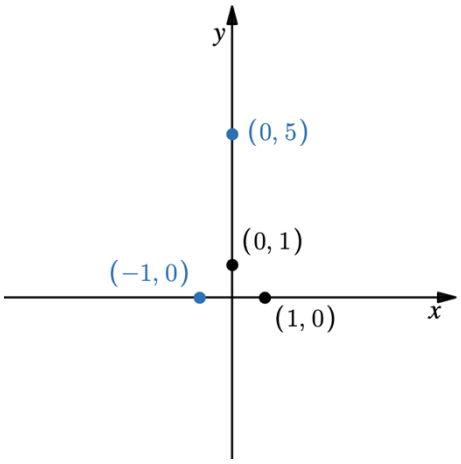
$\mathbf{A}^2 = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \Rightarrow \begin{pmatrix} 4 & p \\ q & p \end{pmatrix}^2 = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \Rightarrow \begin{pmatrix} pq + 16 & p^2 + 4p \\ pq + 4q & p^2 + pq \end{pmatrix} = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix}$

$p^2 + 4p = 0 \Rightarrow p(p+4) = 0 \Rightarrow p = 0$ or $p = -4$

$p \neq 0$ otherwise $pq + 16 \neq 100$

So $p = -4$: $-4q + 16 = 100 \Rightarrow q = -21$

10) a)



10) b)

A rotation, anticlockwise about the origin, by 90°

Followed by

A stretch, parallel to the y-axis, scale factor 5

$\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 5 & 0 \end{pmatrix}$

11) a)

$\begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 40 \end{pmatrix}$

11) b)

$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

11) c)

$$\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}=\begin{pmatrix}0 & -3\\ 1 & 0\end{pmatrix}$$

11) d)

$$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}=\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$$

11) e)

$$\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix}=\begin{pmatrix}-\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{pmatrix}$$

11) f)

$$\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

11) g)

$$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}=\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$$

12)

$$\mathbf{D}=\mathbf{ABC}=\begin{pmatrix}5 & -2\\ 5 & -1\end{pmatrix}\begin{pmatrix}\frac{4}{15} & -\frac{1}{3}\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}$$

$$=\begin{pmatrix}-\frac{2}{3} & \frac{1}{3}\\ \frac{1}{3} & -\frac{2}{3}\end{pmatrix}\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}=\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}$$

So D is a reflection in the line $y=-x$

13)

$$\mathbf{D}=\mathbf{CAB}=\begin{pmatrix}3 & 1\\ 2 & 1\end{pmatrix}\begin{pmatrix}1 & 3\\ -1 & -2\end{pmatrix}\begin{pmatrix}-4 & 7\\ 1 & -2\end{pmatrix}$$

$$=\begin{pmatrix}2 & 7\\ 1 & 4\end{pmatrix}\begin{pmatrix}-4 & 7\\ 1 & -2\end{pmatrix}=\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$$

So D is a rotation by 180° about the origin

14)

$$\begin{pmatrix}\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\cos 225° & -\sin 225°\\ \sin 225° & \cos 225°\end{pmatrix}\begin{pmatrix}\sqrt{2} & -3 & 0\\ 2 & \sqrt{2} & -\sqrt{2}\end{pmatrix}$$

$$=\begin{pmatrix}\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}-\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\end{pmatrix}\begin{pmatrix}\sqrt{2} & -3 & 0\\ 2 & \sqrt{2} & -\sqrt{2}\end{pmatrix}$$

$$=\begin{pmatrix}\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}-1+\sqrt{2} & 1+\frac{3}{2}\sqrt{2} & -1\\ -1-\sqrt{2} & -1+\frac{3}{2}\sqrt{2} & 1\end{pmatrix}$$

$$=\begin{pmatrix}\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\begin{pmatrix}1+\sqrt{2} & 1-\frac{3}{2}\sqrt{2} & -1\\ 1-\sqrt{2} & -1-\frac{3}{2}\sqrt{2} & 1\end{pmatrix}$$

$$=\begin{pmatrix}2+\sqrt{2} & -3+\sqrt{2} & -\sqrt{2}\\ -2+\sqrt{2} & -3-\sqrt{2} & \sqrt{2}\end{pmatrix}$$

So the coordinates of the vertices of triangle X' are $(2+\sqrt{2}, -2+\sqrt{2})$, $(-3+\sqrt{2}, -3-\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$

15)

$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 2\sqrt{2} \\ \sqrt{2} \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} k\cos\theta & -k\sin\theta \\ k\sin\theta & k\cos\theta \end{pmatrix}\begin{pmatrix} 2\sqrt{2} \\ \sqrt{2} \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$

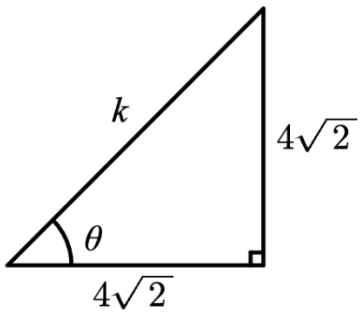
$\Rightarrow \begin{pmatrix} 2\sqrt{2}k\cos\theta - \sqrt{2}k\sin\theta \\ 2\sqrt{2}k\sin\theta + \sqrt{2}k\cos\theta \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$

$2\sqrt{2}k\cos\theta - \sqrt{2}k\sin\theta = -8$ (1)

$2\sqrt{2}k\sin\theta + \sqrt{2}k\cos\theta = -24$ (2)

$2 \times$(1) + (2): $5\sqrt{2}k\cos\theta = -40 \Rightarrow \cos\theta = -\frac{40}{5\sqrt{2}k} = -\frac{4\sqrt{2}}{k}$

(1) - $2 \times$(2): $-5\sqrt{2}k\sin\theta = 40 \Rightarrow \sin\theta = -\frac{40}{5\sqrt{2}k} = -\frac{4\sqrt{2}}{k}$



$k = \sqrt{\left(4\sqrt{2}\right)^2 + \left(4\sqrt{2}\right)^2} = 8$

$\theta = \arctan 1 = 45°$

As both $\cos\theta$ and $\sin\theta$ are negative, $\theta = 225°$

So $k = 8$ and $\theta = 225°$

16)

$\begin{pmatrix} k & 0 \\ 0 & \frac{1}{k} \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} k & 4k \\ \frac{2}{k} & \frac{6}{k} \end{pmatrix}$

So $A'\left(k, \frac{2}{k}\right)$ and $B'\left(4k, \frac{6}{k}\right)$

$145 = (4k - k)^2 + \left(\frac{6}{k} - \frac{2}{k}\right)^2$

$\Rightarrow 145 = (3k)^2 + \left(\frac{4}{k}\right)^2$

$\Rightarrow 145 = 9k^2 + \frac{16}{k^2} \Rightarrow 145k^2 = 9k^4 + 16$

$\Rightarrow 9k^4 - 145k^2 + 16 = 0 \Rightarrow (9k^2 - 1)(k^2 - 16) = 0$

$9k^2 - 1 = 0 \Rightarrow k^2 = \frac{1}{9} \Rightarrow k = \pm\frac{1}{3}$

$k^2 - 16 = 0 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$

As k is a positive integer, we must have $k = 4$

17)

$\mathbf{C} = 3\begin{pmatrix} 2 & -5 \\ -7 & 4 \end{pmatrix} - \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix} = \begin{pmatrix} 6 & -15 \\ -21 & 12 \end{pmatrix} - \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix}$

$= \begin{pmatrix} 0 & 10 \\ 10 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

This is a reflection in the line $y = x$, and an enlargement, centred at the origin, by scale factor 10.

18)

$\mathbf{C} = \begin{pmatrix} 2 & 5 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} -100 & -40 \\ 40 & 20 \end{pmatrix} = \begin{pmatrix} 0 & 20 \\ -20 & 0 \end{pmatrix} = \begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

This is a rotation, anticlockwise about the origin, by $270°$, and an enlargement, centred at the origin, by scale factor 10.

19)

$\mathbf{A}^2 = \begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix}\begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix} = \begin{pmatrix} 4k^2 + km & 2km + 4m \\ 2k^2 + 4k & km + 16 \end{pmatrix}$

$4k^2 + km = 11$ (1)

$km + 16 = 11$ (2)

$\therefore km = -5$

$4k^2 - 5 = 11 \Rightarrow k^2 = 4 \Rightarrow k = \pm 2$

If $k = 2$ then $m = -\frac{5}{2}$

If $k = -2$ then $m = \frac{5}{2}$

$2k^2 + 4k = 0$ (3)

Check in (3):

$k = 2, m = -\frac{5}{2}: 2 \times 2^2 + 4 \times 2 = 16 \neq 0$

$k = -2, m = \frac{5}{2}: 2 \times (-2)^2 + 4 \times (-2) = 0$

$2km + 4m = 0$ (4)

Check in (4):

$k = -2, m = \frac{5}{2}: 2 \times (-2) \times \frac{5}{2} + 4 \times \frac{5}{2} = 0$

So $k = -2$ and $m = \frac{5}{2}$

20)

$\begin{pmatrix} -5 & 4k \\ -2k & p \end{pmatrix}^2 = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix} \Rightarrow \begin{pmatrix} 25-8k^2 & 4kp-20k \\ -2kp+10k & p^2-8k^2 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}$

$25-8k^2 = 9 \Rightarrow 8k^2 = 16 \Rightarrow k^2 = 2 \Rightarrow k = \pm\sqrt{2}$

As k is a positive constant, we must have $k = \sqrt{2}$

$p^2 - 8k^2 = 9 \Rightarrow p^2 - 8\left(\sqrt{2}\right)^2 = 9 \Rightarrow p^2 = 25 \Rightarrow p = \pm 5$

As p is a positive constant, we must have $p = 5$

So $k = \sqrt{2}$ and $p = 5$

21)

$\begin{pmatrix} 3 & -5 \\ 5 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 12 & 0 \\ 0 & 12 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 3a-5c & 3b-5d \\ 5a+c & 5b+d \end{pmatrix} = \begin{pmatrix} 0 & 12 \\ -12 & 0 \end{pmatrix}$

$3a - 5c = 0$ (1)

$5a + c = -12$ (2)

$\therefore a = -\frac{15}{7}$ and $c = -\frac{9}{7}$

$3b - 5d = 12$ (3)

$5b + d = 0$ (4)

$\therefore b = \frac{3}{7}$ and $d = -\frac{15}{7}$

So $a = -\frac{15}{7}$, $b = \frac{3}{7}$, $c = -\frac{9}{7}$ and $d = -\frac{15}{7}$

22) a)

$\mathbf{A}^3 = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}^3 = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}\begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}\begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}$

$= \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}\begin{pmatrix} -54 & 54\sqrt{3} \\ -54\sqrt{3} & -54 \end{pmatrix} = \begin{pmatrix} 648\sqrt{3} & 0 \\ 0 & 648\sqrt{3} \end{pmatrix}$

$\therefore \mathbf{A}^3 = 648\sqrt{3}\mathbf{I}$

22) b)

It is an enlargement, centred at the origin, by scale factor $648\sqrt{3}$

22) c)

It is an enlargement, centred at the origin, by scale factor $\sqrt[3]{648\sqrt{3}} = 6\sqrt{3}$, and a rotation by $\frac{360}{3} = 120°$, anticlockwise about the origin.

23) a)

$$\mathbf{A}^4 = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}^4 = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}$$
$$= \begin{pmatrix} -18 & 0 \\ 0 & -18 \end{pmatrix}\begin{pmatrix} -18 & 0 \\ 0 & -18 \end{pmatrix} = \begin{pmatrix} 324 & 0 \\ 0 & 324 \end{pmatrix}$$
$$\therefore \mathbf{A}^4 = 324\mathbf{I}$$

23) b)

It is an enlargement, centred at the origin, by scale factor 324

23) c)

It is an enlargement, centred at the origin, by scale factor $\sqrt[4]{324} = 3\sqrt{2}$, and a rotation by $\frac{360}{4} = 90°$, anticlockwise about the origin.

23) d)

It is an enlargement, centred at the origin, by scale factor $\frac{1}{3\sqrt{2}}$, and a rotation by $90°$, clockwise about the origin.

24) a)

$$\mathbf{A}^2 = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}^2 = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a^2 - b^2 & -2ab \\ 2ab & a^2 - b^2 \end{pmatrix}$$

24) b)

$$a^2 - b^2 = \left(\frac{1+\sqrt{3}}{4}\right)^2 - \left(\frac{\sqrt{3}-1}{4}\right)^2 = \frac{4+2\sqrt{3}}{16} - \frac{4-2\sqrt{3}}{16} = \frac{4\sqrt{3}}{16} = \frac{\sqrt{3}}{4}$$
$$2ab = 2\left(\frac{1+\sqrt{3}}{4}\right)\left(\frac{\sqrt{3}-1}{4}\right) = 2\left(\frac{2}{16}\right) = \frac{1}{4}$$
$$\mathbf{A}^2 = \begin{pmatrix} \frac{\sqrt{3}}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{\sqrt{3}}{4} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{3}}{4}\times 2 & -\frac{1}{4}\times 2 \\ \frac{1}{4}\times 2 & \frac{\sqrt{3}}{4}\times 2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$
$$\therefore k = \frac{1}{2}$$

24) c) i)

$$\mathbf{A}^6 = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^6 = \begin{pmatrix} 0 & -\frac{1}{8} \\ \frac{1}{8} & 0 \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

This is a rotation, anticlockwise about the origin, by $90°$, and an enlargement, centred at the origin, by scale factor $\frac{1}{8}$

24) c) ii)

$$\mathbf{A}^6 = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^{12} = \begin{pmatrix} -\frac{1}{64} & 0 \\ 0 & -\frac{1}{64} \end{pmatrix} = \frac{1}{64}\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a rotation, about the origin, by 180°, and an enlargement, centred at the origin, by scale factor $\frac{1}{64}$

24) c) iii)

$$\mathbf{A}^{24} = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^{24} = \begin{pmatrix} \frac{1}{4096} & 0 \\ 0 & \frac{1}{4096} \end{pmatrix} = \frac{1}{4096}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This is an enlargement, centred at the origin, by scale factor $\frac{1}{4096}$

24) c) iv)

$$\mathbf{A} = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{\sqrt{2}+\sqrt{6}}{4} & -\frac{\sqrt{6}-\sqrt{2}}{4} \\ \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{2}+\sqrt{6}}{4} \end{pmatrix}$$

This is a rotation, about the origin, by 15°, and an enlargement, centred at the origin, by scale factor $\frac{1}{\sqrt{2}}$

25) a)

T: Reflection in the line $y = \frac{\sqrt{3}}{3}x$:

$\theta = \arctan\frac{\sqrt{3}}{3} = 30°$

Matrix is $\begin{pmatrix} \cos(2\times 30°) & \sin(2\times 30°) \\ \sin(2\times 30°) & -\cos(2\times 30°) \end{pmatrix} = \begin{pmatrix} \cos 60° & \sin 60° \\ \sin 60° & -\cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

U: Reflection in the line $y = \sqrt{3}x$:

$\theta = \arctan\sqrt{3} = 60°$

Matrix is $\begin{pmatrix} \cos(2\times 60°) & \sin(2\times 60°) \\ \sin(2\times 60°) & -\cos(2\times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

T followed by U:

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

25) b)

U followed by T:

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

26) a)

$\cos\theta = 0.28 \Rightarrow \theta = \{73.7°, 286°\}$ (3 SF)

$\sin\theta = 0.96 \Rightarrow \theta = \{73.7°, 106°\}$ (3 SF)

A represents a rotation, anticlockwise about the origin, by 73.7° (3 SF)

26) b)

$\cos 2\theta = -0.8432 \Rightarrow \theta = \{73.7°, 106°\}$ (3 SF)

$\sin 2\theta = 0.5376 \Rightarrow \theta = \{16.3°, 73.7°\}$ (3 SF)

B represents a reflection in the line $y = (\tan 73.7°)x \Rightarrow y = 3.43x$ (3 SF)

26) c)

$$\mathbf{C} = \begin{pmatrix} 0.28 & -0.96 \\ 0.96 & 0.28 \end{pmatrix}\begin{pmatrix} -0.8432 & 0.5376 \\ 0.5376 & 0.8432 \end{pmatrix} = \begin{pmatrix} -0.752192 & -0.658944 \\ -0.658944 & 0.752192 \end{pmatrix}$$

$\cos 2\theta = -0.752192 \Rightarrow \theta = \{69.4°, 111°\}$ (3 SF)

$\sin 2\theta = -0.658944 \Rightarrow \theta = \{111°, 159°\}$ (3 SF)

C represents a reflection in the line $y = (\tan 111°)x \Rightarrow y = -2.61x$ (3 SF)

27)

$y\sqrt{3} - 3x = 0 \Rightarrow y = \frac{3}{\sqrt{3}}x \Rightarrow y = \sqrt{3}x$

$\tan\theta = \sqrt{3} \Rightarrow \theta = \arctan\sqrt{3} = 60°$

$$\therefore \begin{pmatrix} \cos(2\times 60°) & \sin(2\times 60°) \\ \sin(2\times 60°) & -\cos(2\times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

$y\sqrt{3} + x = 0 \Rightarrow y = -\frac{1}{\sqrt{3}}x \Rightarrow y = -\frac{\sqrt{3}}{3}x$

$\tan\theta = -\frac{\sqrt{3}}{3} \Rightarrow \theta = 180° + \arctan\left(-\frac{\sqrt{3}}{3}\right) = 150°$

$$\therefore \begin{pmatrix} \cos(2\times 150°) & \sin(2\times 150°) \\ \sin(2\times 150°) & -\cos(2\times 150°) \end{pmatrix} = \begin{pmatrix} \cos 300° & \sin 300° \\ \sin 300° & -\cos 300° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a reflection in the line $y = -x$

28) a)

$$\mathbf{A}^2 = \begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix}^2 = \begin{pmatrix} 18 & 0 \\ 0 & 18 \end{pmatrix}$$

It is an enlargement, centred at the origin, by scale factor 18

28) b)

It is an enlargement, centred at the origin, by scale factor $\sqrt{18} = 3\sqrt{2}$

$$\frac{1}{3\sqrt{2}}\mathbf{A} = \frac{1}{3\sqrt{2}}\begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$\cos 2\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{22.5°, 157.5°\}$

$\sin 2\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{22.5°, 67.5°\}$

So it is a reflection in the line $y = (\tan 22.5°)x \Rightarrow y = 0.414x$ (3 SF)

29)

$\begin{pmatrix} -\cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\sqrt{2} \end{pmatrix} \Rightarrow \begin{pmatrix} -\cos 2\theta + \sin 2\theta \\ -\sin 2\theta - \cos 2\theta \end{pmatrix} = \begin{pmatrix} 0 \\ -\sqrt{2} \end{pmatrix}$

$-\cos 2\theta + \sin 2\theta = 0 \Rightarrow \sin 2\theta = \cos 2\theta \Rightarrow \frac{\sin 2\theta}{\cos 2\theta} = 1 \Rightarrow \tan 2\theta = 1$

$\Rightarrow \theta = \{22.5°, 112.5°, 202.5°, 292.5°\}$

If $\theta = 22.5°$, then $-\sin(2 \times 22.5°) - \cos(2 \times 22.5°) = -\sin 45° - \cos 45°$

$= -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$

So $\theta = 22.5°$ is the smallest value of θ

30)

$\begin{pmatrix} 3\cos\theta & -\sin\theta \\ \sin\theta & 3\cos\theta \end{pmatrix}\begin{pmatrix} 3 \\ -\frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 9\cos\theta + \frac{1}{\sqrt{3}}\sin\theta \\ 3\sin\theta - \frac{3}{\sqrt{3}}\cos\theta \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 9\cos\theta + \frac{\sqrt{3}}{3}\sin\theta \\ 3\sin\theta - \sqrt{3}\cos\theta \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix}$

$3\sin\theta - \sqrt{3}\cos\theta = 0 \Rightarrow 3\sin\theta = \sqrt{3}\cos\theta \Rightarrow \frac{3\sin\theta}{\cos\theta} = \sqrt{3} \Rightarrow 3\tan\theta = \sqrt{3}$

$\Rightarrow \tan\theta = \frac{\sqrt{3}}{3} \Rightarrow \theta = 30°$ as θ is a positive acute angle

$k = 9\cos 30° + \frac{\sqrt{3}}{3}\sin 30° = \frac{9\sqrt{3}}{2} + \frac{\sqrt{3}}{6} = \frac{14\sqrt{3}}{3}$

31) a)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

So it is a rotation by $40°$, anticlockwise about the origin

31) b)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$\cos\theta = \cos 40° \Rightarrow \theta = \{40°, 320°\}$

$\sin\theta = -\sin 40° \Rightarrow \theta = \{220°, 320°\}$

So it is a rotation by $320°$, anticlockwise about the origin

Or a rotation by $40°$, clockwise about the origin

31) c)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

So it is a reflection in the line $y = (\tan 20°)x \Rightarrow y = 0.364x$ (3 SF)

31) d)

This is similar to the general reflection matrix $\begin{pmatrix}\cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta\end{pmatrix}$

$\cos 2\theta = -\cos 40° \Rightarrow \theta = \{70°, 110°\}$

$\sin 2\theta = \sin 40° \Rightarrow \theta = \{20°, 70°\}$

So it is a reflection in the line $y = (\tan 70°)x \Rightarrow y = 2.75x$ (3 SF)

32) a)

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$

$\cos\theta = -\cos 40° \Rightarrow \theta = \{140°, 220°\}$

$\sin\theta = \sin 40° \Rightarrow \theta = \{40°, 140°\}$

So it is a rotation by $140°$, anticlockwise about the origin

32) b)

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$

$\cos\theta = -\cos 40° \Rightarrow \theta = \{140°, 220°\}$

$\sin\theta = -\sin 40° \Rightarrow \theta = \{220°, 320°\}$

So it is a rotation by $220°$, anticlockwise about the origin

32) c)

This is similar to the general reflection matrix $\begin{pmatrix}\cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta\end{pmatrix}$

$\cos 2\theta = \cos 40° \Rightarrow \theta = \{20°, 160°\}$

$\sin 2\theta = -\sin 40° \Rightarrow \theta = \{110°, 160°\}$

So it is a reflection in the line $y = (\tan 160°)x \Rightarrow y = -0.364x$ (3 SF)

32) d)

This is similar to the general reflection matrix $\begin{pmatrix}\cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta\end{pmatrix}$

$\cos 2\theta = -\cos 40° \Rightarrow \theta = \{70°, 110°\}$

$\sin 2\theta = -\sin 40° \Rightarrow \theta = \{110°, 160°\}$

So it is a reflection in the line $y = (\tan 110°)x \Rightarrow y = -2.75x$ (3 SF)

33) a)

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$

So it is a rotation by $140°$, anticlockwise about the origin

33) b)

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$

$\cos\theta = \cos 140° \Rightarrow \theta = \{140°, 220°\}$

$\sin\theta = -\sin 140° \Rightarrow \theta = \{220°, 320°\}$

So it is a rotation by 220°, anticlockwise about the origin

33) c)

This is similar to the general reflection matrix $\begin{pmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{pmatrix}$

So it is a reflection in the line $y = (\tan 70°)x \Rightarrow y = 2.75x$ (3 SF)

33) d)

This is similar to the general reflection matrix $\begin{pmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{pmatrix}$

$\cos 2\theta = -\cos 140° \Rightarrow \theta = \{20°, 160°\}$

$\sin 2\theta = \sin 140° \Rightarrow \theta = \{20°, 70°\}$

So it is a reflection in the line $y = (\tan 20°)x \Rightarrow y = 0.364x$ (3 SF)

34) a)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$

So $\begin{pmatrix}\sin 40° & -\cos 40°\\ \cos 40° & \sin 40°\end{pmatrix} = \begin{pmatrix}\cos 50° & -\sin 50°\\ \sin 50° & \cos 50°\end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$

So it is a rotation by 50°, anticlockwise about the origin

34) b)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$

So $\begin{pmatrix}\sin 40° & \cos 40°\\ -\cos 40° & \sin 40°\end{pmatrix} = \begin{pmatrix}\cos 50° & \sin 50°\\ -\sin 50° & \cos 50°\end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$

$\cos\theta = \cos 50° \Rightarrow \theta = \{50°, 310°\}$

$\sin\theta = -\sin 50° \Rightarrow \theta = \{230°, 310°\}$

So it is a rotation by 310°, anticlockwise about the origin

Or a rotation by 50°, clockwise about the origin

34) c)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$

So $\begin{pmatrix} \sin 40° & \cos 40° \\ \cos 40° & -\sin 40° \end{pmatrix} = \begin{pmatrix} \cos 50° & \sin 50° \\ \sin 50° & -\cos 50° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

So it is a reflection in the line $y = (\tan 25°)x \Rightarrow y = 0.466x$ (3 SF)

34) d)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$

So $\begin{pmatrix} -\sin 40° & \cos 40° \\ \cos 40° & \sin 40° \end{pmatrix} = \begin{pmatrix} -\cos 50° & \sin 50° \\ \sin 50° & \cos 50° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

$\cos 2\theta = -\cos 50° \Rightarrow \theta = \{65°, 115°\}$

$\sin 2\theta = \sin 50° \Rightarrow \theta = \{25°, 65°\}$

So it is a reflection in the line $y = (\tan 65°)x \Rightarrow y = 2.14x$ (3 SF)

35) a)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$

So $\begin{pmatrix} \sin 140° & -\cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} \cos 310° & -\sin 310° \\ \sin 310° & \cos 310° \end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

So it is a rotation by $310°$, anticlockwise about the origin

Or a rotation by $50°$, clockwise about the origin

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$

So $\begin{pmatrix} \sin 140° & -\cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} \cos 50° & \sin 50° \\ -\sin 50° & \cos 50° \end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$\cos\theta = \cos 50° \Rightarrow \theta = \{50°, 310°\}$

$\sin\theta = -\sin 50° \Rightarrow \theta = \{230°, 310°\}$

So it is a rotation by $310°$, anticlockwise about the origin

Or a rotation by $50°$, clockwise about the origin

35) b)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$

So $\begin{pmatrix} \sin 140° & \cos 140° \\ -\cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} \cos 310° & \sin 310° \\ -\sin 310° & \cos 310° \end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$\cos\theta = \cos 310° \Rightarrow \theta = \{50°, 310°\}$

$\sin\theta = -\sin 310° \Rightarrow \theta = \{50°, 130°\}$

So it is a rotation by $50°$, anticlockwise about the origin

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$

So $\begin{pmatrix} \sin 140° & \cos 140° \\ -\cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} \cos 50° & -\sin 50° \\ \sin 50° & \cos 50° \end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

So it is a rotation by $50°$, anticlockwise about the origin

Or a rotation by $310°$, anticlockwise about the origin

35) c)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$

So $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix} = \begin{pmatrix} \cos 310° & \sin 310° \\ \sin 310° & -\cos 310° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

So it is a reflection in the line $y = (\tan 155°)x$, which is roughly $y = -0.466x$

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$

So $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix} = \begin{pmatrix} \cos 50° & -\sin 50° \\ -\sin 50° & -\cos 50° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

$\cos 2\theta = \cos 50° \Rightarrow \theta = \{25°, 155°, 205°, 335°\}$

$\sin 2\theta = -\sin 50° \Rightarrow \theta = \{115°, 155°, 295°, 335°\}$

So it is a reflection in the line $y = (\tan 155°)x$, which is roughly $y = -0.466x$

35) d)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$

So $\begin{pmatrix} -\sin 140° & \cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} -\cos 310° & \sin 310° \\ \sin 310° & \cos 310° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

$\cos 2\theta = -\cos 310° \Rightarrow \theta = \{65°, 115°\}$

$\sin 2\theta = \sin 310° \Rightarrow \theta = \{115°, 155°\}$

So it is a reflection in the line $y = (\tan 115°)x$, which is roughly $y = -2.14x$

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$

So $\begin{pmatrix} -\sin 140° & \cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} -\cos 50° & -\sin 50° \\ -\sin 50° & \cos 50° \end{pmatrix}$

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$

$\cos 2\theta = -\cos 50° \Rightarrow \theta = \{65°, 115°, 245°, 295°\}$

$\sin 2\theta = -\sin 50° \Rightarrow \theta = \{115°, 155°, 295°, 335°\}$

So it is a reflection in the line $y = (\tan 115°)x$, which is roughly $y = -0.466x$

36)

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix}^2 = \begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} \Rightarrow \begin{pmatrix} p^2 - q^2 & -2pq \\ 2pq & p^2 - q^2 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

$2pq = \frac{1}{2} \Rightarrow q = \frac{1}{4p}$

$p^2 - q^2 = -\frac{\sqrt{3}}{2} \Rightarrow p^2 - \left(\frac{1}{4p}\right)^2 = -\frac{\sqrt{3}}{2} \Rightarrow p^2 - \frac{1}{16p^2} = -\frac{\sqrt{3}}{2}$

$\Rightarrow 16p^4 - 1 = -8\sqrt{3}p^2$

$\Rightarrow 16p^4 + 8\sqrt{3}p^2 - 1 = 0$

$\Rightarrow p^2 = \frac{-8\sqrt{3} \pm \sqrt{(8\sqrt{3})^2 - 4 \times 16 \times (-1)}}{32}$

$\Rightarrow p^2 = \frac{-8\sqrt{3} \pm 16}{32} \Rightarrow p^2 = \frac{-\sqrt{3} \pm 2}{4}$

$-\sqrt{3} - 2 < 0$ so $p^2 = \frac{-\sqrt{3}+2}{4} \Rightarrow p = \sqrt{\frac{2-\sqrt{3}}{4}}$

$\begin{pmatrix} \cos 75° & -\sin 75° \\ \sin 75° & \cos 75° \end{pmatrix}^2 = \begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix}$

$\therefore \cos 75° = \sqrt{\frac{2-\sqrt{3}}{4}}$

37)

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix}^2 = \begin{pmatrix} \cos 105° & -\sin 105° \\ \sin 105° & \cos 105° \end{pmatrix} \Rightarrow \begin{pmatrix} p^2-q^2 & -2pq \\ 2pq & p^2-q^2 \end{pmatrix} = \begin{pmatrix} \frac{-\sqrt{6}+\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} \\ \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{-\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$$

$2pq = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow p = \frac{\sqrt{6}+\sqrt{2}}{8q}$

$p^2 - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4} \Rightarrow \left(\frac{\sqrt{6}+\sqrt{2}}{8q}\right)^2 - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4} \Rightarrow \frac{8+4\sqrt{3}}{64q^2} - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4}$

$\Rightarrow 8 + 4\sqrt{3} - 64q^4 = 16\left(-\sqrt{6}+\sqrt{2}\right)q^2$

$\Rightarrow 64q^4 + 16\left(-\sqrt{6}+\sqrt{2}\right)q^2 - 4\sqrt{3} - 8 = 0$

$\Rightarrow 16q^4 + 4\left(-\sqrt{6}+\sqrt{2}\right)q^2 - \sqrt{3} - 2 = 0$

$\Rightarrow q^2 = \frac{-4\left(-\sqrt{6}+\sqrt{2}\right) \pm \sqrt{\left(-4\left(-\sqrt{6}+\sqrt{2}\right)\right)^2 - 4\times 16\times\left(-\sqrt{3}-2\right)}}{32}$

$\Rightarrow q^2 = \frac{-4\left(-\sqrt{6}+\sqrt{2}\right)\pm 16}{32} \Rightarrow q^2 = \frac{\sqrt{6}-\sqrt{2}\pm 4}{8}$

$\sqrt{6}-\sqrt{2}-4 < 0$ so $q^2 = \frac{\sqrt{6}-\sqrt{2}+4}{8} \Rightarrow q = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$

$$\begin{pmatrix} \cos 52.5° & -\sin 52.5° \\ \sin 52.5° & \cos 52.5° \end{pmatrix}^2 = \begin{pmatrix} \cos 105° & -\sin 105° \\ \sin 105° & \cos 105° \end{pmatrix}$$

$\therefore \sin 52.5° = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$

38) a)

$\det \mathbf{A} = 10 \times 10 - (-10) \times 10 = 200$

$\sqrt{200} = 10\sqrt{2}$

$\begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 10\sqrt{2} \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix} \Rightarrow \begin{pmatrix} 10\sqrt{2}a & 10\sqrt{2}b \\ 10\sqrt{2}c & 10\sqrt{2}d \end{pmatrix} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix}$

$10\sqrt{2}a = 10 \Rightarrow a = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$10\sqrt{2}b = -10 \Rightarrow b = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$

$10\sqrt{2}c = 10 \Rightarrow c = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$10\sqrt{2}d = 10 \Rightarrow d = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

So $\mathbf{A} = \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 10\sqrt{2} \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $10\sqrt{2}$, and a Rotation, anticlockwise about the origin, by scale factor $45°$ (either order).

38) b)

$\det \mathbf{A} = -8\sqrt{3} \times 8\sqrt{3} - 0 \times 0 = -192$

$\sqrt{192} = 8\sqrt{3}$

$\begin{pmatrix} 8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 8\sqrt{3}a & 8\sqrt{3}b \\ 8\sqrt{3}c & 8\sqrt{3}d \end{pmatrix} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}$

$8\sqrt{3}a = -8\sqrt{3} \Rightarrow a = -1$

$8\sqrt{3}b = 0 \Rightarrow b = 0$

$8\sqrt{3}c = 0 \Rightarrow c = 0$

$8\sqrt{3}d = 8\sqrt{3} \Rightarrow d = 1$

So $\mathbf{A} = \begin{pmatrix} 8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $8\sqrt{3}$, and a Reflection in the y-axis (either order).

38) c)

$\det \mathbf{A} = 10\sqrt{3} \times 10\sqrt{3} - (-10) \times 10 = 400$

$\sqrt{400} = 20$

$\begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 20a & 20b \\ 20c & 20d \end{pmatrix} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix}$

$20a = 10\sqrt{3} \Rightarrow a = \frac{\sqrt{3}}{2}$

$20b = -10 \Rightarrow b = -\frac{1}{2}$

$20c = 10 \Rightarrow c = \frac{1}{2}$

$20d = 10\sqrt{3} \Rightarrow d = \frac{\sqrt{3}}{2}$

So $\mathbf{A} = \begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix}\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\}$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor 20, and a Rotation, anticlockwise about the origin, by scale factor 30° (either order).

38) d)

$\det \mathbf{A} = \sqrt{5} \times \sqrt{5} - \sqrt{5} \times (-\sqrt{5}) = 10$

$\begin{pmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{10} \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix} \Rightarrow \begin{pmatrix} \sqrt{10}a & \sqrt{10}b \\ \sqrt{10}c & \sqrt{10}d \end{pmatrix} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix}$

$\sqrt{10}a = \sqrt{5} \Rightarrow a = \frac{\sqrt{2}}{2}$

$\sqrt{10}b = \sqrt{5} \Rightarrow b = \frac{\sqrt{2}}{2}$

$\sqrt{10}c = -\sqrt{5} \Rightarrow c = -\frac{\sqrt{2}}{2}$

$\sqrt{10}d = \sqrt{5} \Rightarrow d = \frac{\sqrt{2}}{2}$

So $\mathbf{A} = \begin{pmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{10} \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{225°, 315°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $\sqrt{10}$, and a Rotation, anticlockwise about the origin, by 315° (either order).

38) e)

$\det \mathbf{A} = 5\sqrt{30} \times 5\sqrt{30} - 5\sqrt{10} \times \left(-5\sqrt{10}\right) = 1000$

$\begin{pmatrix} 10\sqrt{10} & 0 \\ 0 & 10\sqrt{10} \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 10\sqrt{10}a & 10\sqrt{10}b \\ 10\sqrt{10}c & 10\sqrt{10}d \end{pmatrix} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$

$10\sqrt{10}a = 5\sqrt{30} \Rightarrow a = \frac{\sqrt{3}}{2}$

$10\sqrt{10}b = 5\sqrt{10} \Rightarrow b = \frac{1}{2}$

$10\sqrt{10}c = -5\sqrt{10} \Rightarrow c = -\frac{1}{2}$

$10\sqrt{10}d = 5\sqrt{30} \Rightarrow d = \frac{\sqrt{3}}{2}$

So $\mathbf{A} = \begin{pmatrix} 10\sqrt{10} & 0 \\ 0 & 10\sqrt{10} \end{pmatrix}\begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\}$

$\sin\theta = -\frac{1}{2} \Rightarrow \theta = \{210°, 330°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $10\sqrt{10}$, and a Rotation, anticlockwise about the origin, by $330°$ (either order).

38) f)

$\det \mathbf{A} = (12 - 4\sqrt{3}) \times (12 - 4\sqrt{3}) - (12 + 4\sqrt{3}) \times (-12 - 4\sqrt{3})$

$= (192 - 96\sqrt{3}) - (-192 - 96\sqrt{3}) = 384$

$\sqrt{384} = 8\sqrt{6}$

$\begin{pmatrix} 8\sqrt{6} & 0 \\ 0 & 8\sqrt{6} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 12 - 4\sqrt{3} & 12 + 4\sqrt{3} \\ -12 - 4\sqrt{3} & 12 - 4\sqrt{3} \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 8\sqrt{6}a & 8\sqrt{6}b \\ 8\sqrt{6}c & 8\sqrt{6}d \end{pmatrix} = \begin{pmatrix} 12 - 4\sqrt{3} & 12 + 4\sqrt{3} \\ -12 - 4\sqrt{3} & 12 - 4\sqrt{3} \end{pmatrix}$

$8\sqrt{6}a = 12 - 4\sqrt{3} \Rightarrow a = \frac{\sqrt{6}-\sqrt{2}}{4}$

$8\sqrt{6}b = 12 + 4\sqrt{3} \Rightarrow b = \frac{\sqrt{6}+\sqrt{2}}{4}$

$8\sqrt{6}c = -12 - 4\sqrt{3} \Rightarrow c = -\frac{\sqrt{6}+\sqrt{2}}{4}$

$8\sqrt{6}d = 12 - 4\sqrt{3} \Rightarrow d = \frac{\sqrt{6}-\sqrt{2}}{4}$

So $\mathbf{A} = \begin{pmatrix} 8\sqrt{6} & 0 \\ 0 & 8\sqrt{6} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ -\frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$

$\sin\theta = -\frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{255°, 285°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $8\sqrt{6}$, and a Rotation, anticlockwise about the origin, by 285° (either order).

39)

$\mathbf{A} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -k \\ -k & 0 \end{pmatrix}$

$|A| = -12 = 0 \times 0 - (-k) \times (-k)$

$\Rightarrow k^2 = 12 \Rightarrow k = \sqrt{12} = 2\sqrt{3}$

$\therefore \mathbf{A} = \begin{pmatrix} 0 & -2\sqrt{3} \\ -2\sqrt{3} & 0 \end{pmatrix}$

40)

$$\begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{pmatrix}\begin{pmatrix} -\sqrt{3} & -4\sqrt{3} & -3\sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -6\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{pmatrix}\begin{pmatrix} -\sqrt{3} & -4\sqrt{3} & -3\sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -6\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -4 & \frac{15}{2} \\ -\frac{5\sqrt{3}}{2} & -4\sqrt{3} & -\frac{27\sqrt{3}}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5\sqrt{3}}{2} & -20\sqrt{3} & \frac{75\sqrt{3}}{2} \\ -\frac{5\sqrt{3}}{2} & -4\sqrt{3} & -\frac{27\sqrt{3}}{2} \end{pmatrix}$$

So the coordinates of the vertices of triangle X' are $\left(\frac{5\sqrt{3}}{2}, -\frac{5\sqrt{3}}{2}\right)$, $\left(-20\sqrt{3}, -4\sqrt{3}\right)$ and $\left(\frac{75\sqrt{3}}{2}, -\frac{27\sqrt{3}}{2}\right)$

Area of X is $\frac{1}{2} \times \left(-\sqrt{3} - \left(-4\sqrt{3}\right)\right) \times \left(-\sqrt{2} - \left(-6\sqrt{2}\right)\right) = \frac{1}{2} \times 3\sqrt{3} \times 5\sqrt{2} = \frac{15\sqrt{6}}{2}$ units^2

Area of X' is $\frac{15\sqrt{6}}{2} \times \begin{vmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{vmatrix} = \frac{15\sqrt{6}}{2} \times 5\sqrt{3} \times 2\sqrt{2} = 450$ units^2

41)

$|\mathbf{A}| = \begin{vmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{vmatrix} = 3 \times (-3) - (-3\sqrt{3}) \times (-3\sqrt{3}) = -9 - 27 = -36$

So $|\mathbf{R}| = 36$, $|\mathbf{P}| = 1$ (rotations maintain area and orientation), $|\mathbf{Q}| = -1$ (reflections maintain area but reverse orientation)

$\therefore \mathbf{R} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$ so $\mathbf{R}$ represents an enlargement centred at the origin by scale factor 6

$$\begin{pmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{6} \end{pmatrix}\begin{pmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

So $\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ is a composition of a reflection and a rotation:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \text{ where } ad - bc = -1$$

$$\Rightarrow \begin{pmatrix} a\cos\theta + b\sin\theta & -a\sin\theta + b\cos\theta \\ c\cos\theta + d\sin\theta & -c\sin\theta + d\cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$a\cos\theta + b\sin\theta = \frac{1}{2}$ (1)

$-a\sin\theta + b\cos\theta = -\frac{\sqrt{3}}{2}$ (2)

$c\cos\theta + d\sin\theta = -\frac{\sqrt{3}}{2}$ (3)

$-c\sin\theta + d\cos\theta = -\frac{1}{2}$ (4)

(1)&(4):

$a\cos\theta + b\sin\theta = c\sin\theta - d\cos\theta$

$\Rightarrow b = c$ and $a = -d$

(2)&(3):

$-a\sin\theta + b\cos\theta = c\cos\theta + d\sin\theta$

$\Rightarrow -a = d$ and $b = c$

As $ad - bc = -1$, we must have that $a(-a) - b(b) = -1 \Rightarrow -a^2 - b^2 = -1$

$\Rightarrow a^2 + b^2 = 1$

Let $a = 1$ and $b = 0$, then $c = 0$ and $d = -1$

$\therefore \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ so $\mathbf{Q}$ represents a reflection in the x-axis

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = \{60°, 300°\}$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60°, 120°\}$

$\therefore \mathbf{P} = \begin{pmatrix} \cos 60° & -\sin 60° \\ \sin 60° & \cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$ so $\mathbf{P}$ represents a rotation by 60°, anticlockwise about the origin.

$$\therefore \mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

Alternative Method:

Let $a = 0$ and $b = 1$, then $c = 1$ and $d = 0$

$\therefore \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ so $\mathbf{Q}$ represents a reflection in the line $y = x$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$\cos\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{150°, 210°\}$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$

$\therefore \mathbf{P} = \begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$ so $\mathbf{P}$ represents a rotation by 150°, anticlockwise about the origin.

$$\therefore \mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

So $\mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$ or $\mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$

42)

Let $\mathbf{A} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$

Consider $\mathbf{AB}$:

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$$

$a = -\frac{1}{5}$ and $b = 5$

Check: $1 + ab = 1 + \left(-\frac{1}{5}\right) \times 5 = 0$

If we had instead considered $\mathbf{BA}$:

$$\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & a \\ b & ab+1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$$

We would then have had $1 = 0$, so $\mathbf{BA}$ is not the correct order.

So $\mathbf{A} = \begin{pmatrix} 1 & -\frac{1}{5} \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$

43) a)

Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(-5, 1)$

43) b)

Let $\mathbf{C} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\mathbf{CA} = \mathbf{B} \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \sqrt{2} & -6\sqrt{2} \\ \sqrt{2} & -4\sqrt{2} \end{pmatrix}$

$\Rightarrow \begin{pmatrix} a & b-5a \\ c & d-5c \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -3\sqrt{2} \\ \frac{\sqrt{2}}{2} & -2\sqrt{2} \end{pmatrix}$

$a = \frac{\sqrt{2}}{2}$ and $c = \frac{\sqrt{2}}{2}$

$b - 5 \times \frac{\sqrt{2}}{2} = -3\sqrt{2} \Rightarrow b = -\frac{\sqrt{2}}{2}$

$d - 5 \times \frac{\sqrt{2}}{2} = -2\sqrt{2} \Rightarrow d = \frac{\sqrt{2}}{2}$

$\therefore \mathbf{C} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

So it is a rotation by $45°$, anticlockwise about the origin

44)

Let $\mathbf{A} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$

$\mathbf{ABC} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & c \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} a & -1+ac \\ 1 & c \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$

$\therefore a = 2$ and $c = 4$

Check: $-1 + ac = -1 + 2 \times 4 = 7$

So $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$

45)

Area of triangle $X = \frac{1}{2} \times 3 \times 2 = 3$ units^2

Area of triangle $X' = \frac{1}{2} \times 3 \times 4 = 6$ units^2

$\frac{6}{3} = 2$ so the matrix that represents the stretch is either $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

As $B(3,0)$ is mapped to $B'(0,3)$, this can be achieved via a rotation, anticlockwise about the origin, by 90°, followed by a stretch parallel to the x-axis, scale factor 2, followed by a shear that fixes the y-axis. The rotation matrix would be $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. If the stretch is parallel to the x-axis then we have $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

So far we have:

$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Then:

$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 3 & 3 \\ 2 & 0 & 2 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2 & 0 & -2 \\ 0 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} -4 & 0 & -4 \\ 0 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -4 & 0 & -4 \\ -4k & 3 & -4k+3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$

$-4k = 12 \Rightarrow k = -3$

$-4k + 3 = 15 \Rightarrow k = -3$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$